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Global Earthquake Risk Analytics using R

Exploratory analysis, risk scoring, clustering, machine learning
and an interactive Shiny dashboard

Case study report

B.E. (Artificial Intelligence and Data Science)

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Source code and data:
https://github.com/chaithanyakumarreddy57-spec/Global-Earthquake-Risk-Analytics

Abstract

This case study analyses 23,232 natural earthquakes of magnitude 5.5 and above recorded worldwide between 1965 and 2016, using R. After cleaning the catalogue (23,412 raw records, from which 180 explosions and rock bursts were removed), the study explores magnitude, depth, time and geography, defines a transparent 0–100 *risk score* that combines magnitude (60%) and shallowness (40%), and groups the earthquakes into four risk levels. About 34.3% of the events fall in the High or Severe levels. K-means clustering with six clusters is applied to position, depth and magnitude, and three classifiers (logistic regression, decision tree, random forest) are trained to recognise High/Severe events from location, time and catalogue information, without using magnitude or depth directly. The random forest performed best on the held-out test set (accuracy 0.69, F1 0.43, AUC 0.72), but the margin over a majority-class baseline (0.66 accuracy) is modest, which is discussed openly in the limitations. The results are presented in an interactive Shiny dashboard. All code, data and outputs are in the public GitHub repository.

Keywords: earthquakes, risk scoring, K-means, random forest, R, Shiny.

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1 Introduction and problem statement

Earthquakes are among the most damaging natural hazards. Their effects depend on how large an event is, how deep it occurs, and where it happens. Large, shallow earthquakes release energy close to the surface and are generally the most dangerous. Decision makers and students who want to understand global patterns need a clear way to explore large catalogues of events and to rank them by potential severity.

This project turns a public catalogue of significant earthquakes into an understandable picture of global seismic activity. It asks four questions:

1. Where and when do significant earthquakes occur, and how are their magnitude and depth distributed?
2. Can magnitude and depth be combined into a simple, explainable severity score?
3. Do earthquakes fall into natural groups?
4. How well can machine-learning models recognise High/Severe events from location, time and catalogue information alone?

Scope. The score and models describe and classify *recorded events*. They are not a forecast of future earthquakes and they do not account for population, buildings or local geology, so they do not measure human or economic risk.

2 Objectives and dataset

2.1 Objectives

- Clean and document a global earthquake catalogue reproducibly in R.
- Explore magnitude, depth, temporal and geographic patterns.
- Define a risk score and four risk categories, and map them.
- Cluster earthquakes with K-means and interpret the groups.
- Train and compare three classification models with appropriate metrics.
- Deliver an interactive Shiny dashboard and a reproducible public repository.

2.2 Dataset

The raw file `data/raw/database.csv` contains 23,412 records and 21 columns with earthquakes of magnitude 5.5 or higher from January 1965 to December 2016. Its structure matches the publicly available *Significant Earthquakes, 1965–2016* dataset (USGS catalogue, distributed on Kaggle). The original source link is given in the README and the references. Important variables are listed in Table 1.

Table 1: Main variables of the raw dataset.

Variable	Meaning
Date, Time	Date and UTC time of the event
Latitude, Longitude	Epicentre in decimal degrees
Depth	Depth of the focus in km
Magnitude	Earthquake magnitude (5.5 to 9.1)
Magnitude Type	Scale used (e.g. MW, MB, MS)
Type	Earthquake, Nuclear Explosion, Explosion, Rock Burst

Status	Automatic or Reviewed solution
ID, Source, ...	Catalogue identifiers and data sources

3 Methodology and workflow

The pipeline in `earthquake_analysis.R` runs from top to bottom and writes every table, figure and model used in this report. Figure 1 shows the workflow.

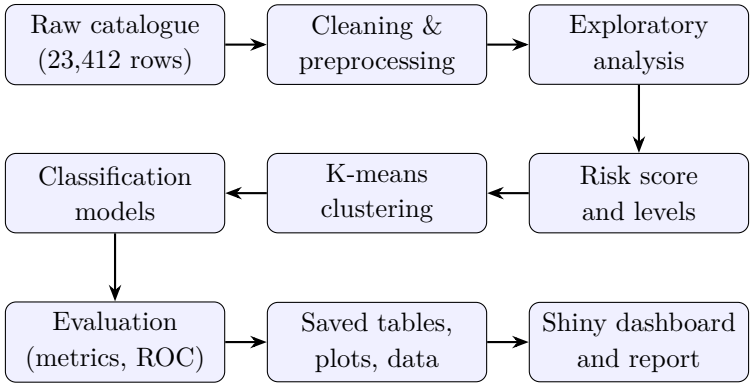


Figure 1: Project workflow from raw data to dashboard and report.

The main R packages are `tidyverse`, `lubridate`, `janitor`, `ggplot2`, `leaflet`, `maps`, `caret`, `rpart`, `randomForest`, `pROC`, `cluster`, `factoextra`, `shiny`, `bslib`, `plotly` and `DT`. A random seed (`set.seed(42)`) is set at the start of the analysis.

4 Data cleaning and preprocessing

The cleaning steps and what they changed are summarised in Table 2.

Table 2: Data cleaning steps and their effect.

Step	Action
Column names	Converted to snake_case with janitor
Duplicates	0 duplicate rows and 0 duplicate IDs found; none removed
Non-earthquake events	180 records removed (4 explosion, 175 nuclear explosion, 1 rock burst)
Columns with >60% missing	7 columns dropped (magnitude_error, horizontal_error, horizontal_distance, magnitude_seismic_stations, depth_error, depth_seismic_stations, azimuthal_gap)
Remaining missing values	Magnitude type: 3 missing set to 'Unknown'; root-mean-square error filled with the median
Negative depths	3 negative depths (minimum -1.1 km) set to 0 km
Date and time	3 rows stored as ISO timestamps parsed separately; all dates parsed
Outliers	IQR rule flags 1,471 magnitude and 3,239 depth values; kept because large and deep events are real

After cleaning the dataset holds 23,232 earthquakes. The following features were then created: year, month, hour; a magnitude category (Moderate 5.5–5.9, Strong 6.0–6.9, Major 7.0–7.9, Great 8.0+); a depth category (Shallow below 70 km, Intermediate 70–300 km, Deep above 300 km); log depth; encoded categories; and z-score scaled latitude, longitude, depth and magnitude.

Rare magnitude types were grouped into “Other” so that train and test sets never contain unseen levels. The final cleaned file has 35 columns, including the risk score, risk level, region and cluster added later in the pipeline.

5 Exploratory data analysis

5.1 Magnitude and depth

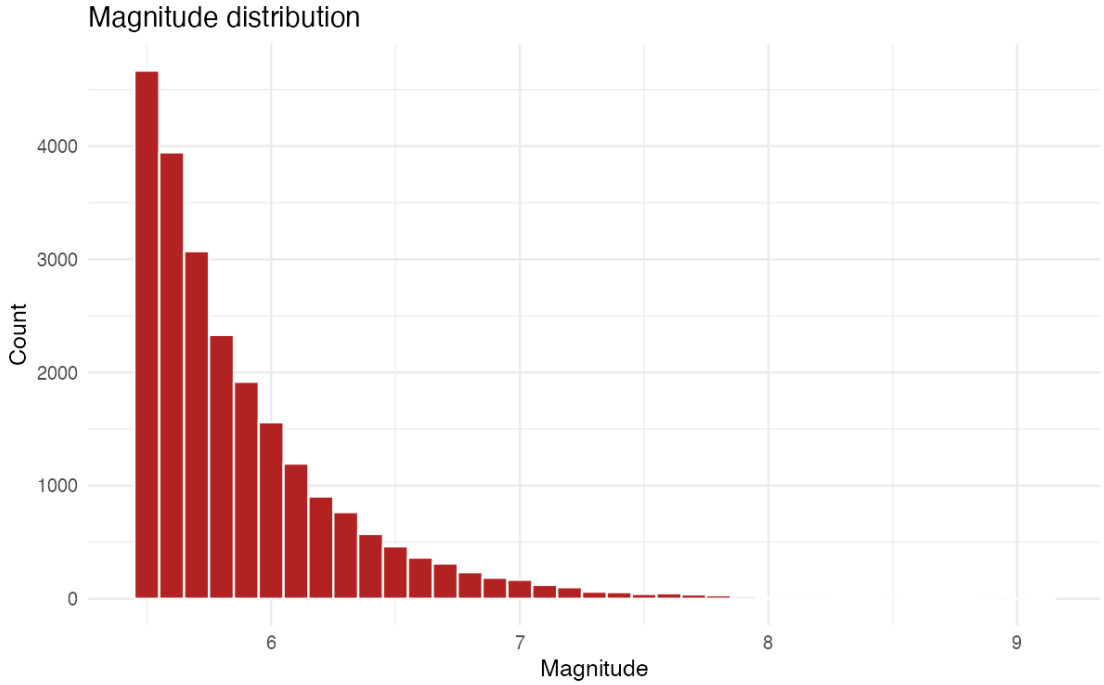


Figure 2: Distribution of earthquake magnitude (bin width 0.1).

Magnitudes are strongly right-skewed (histogram above). The median magnitude is 5.70 and the mean 5.88. Because the dataset starts at 5.5, the count falls quickly with magnitude. Only 3.2% of events reach magnitude 7 or more, and 40 reach magnitude 8 or more.

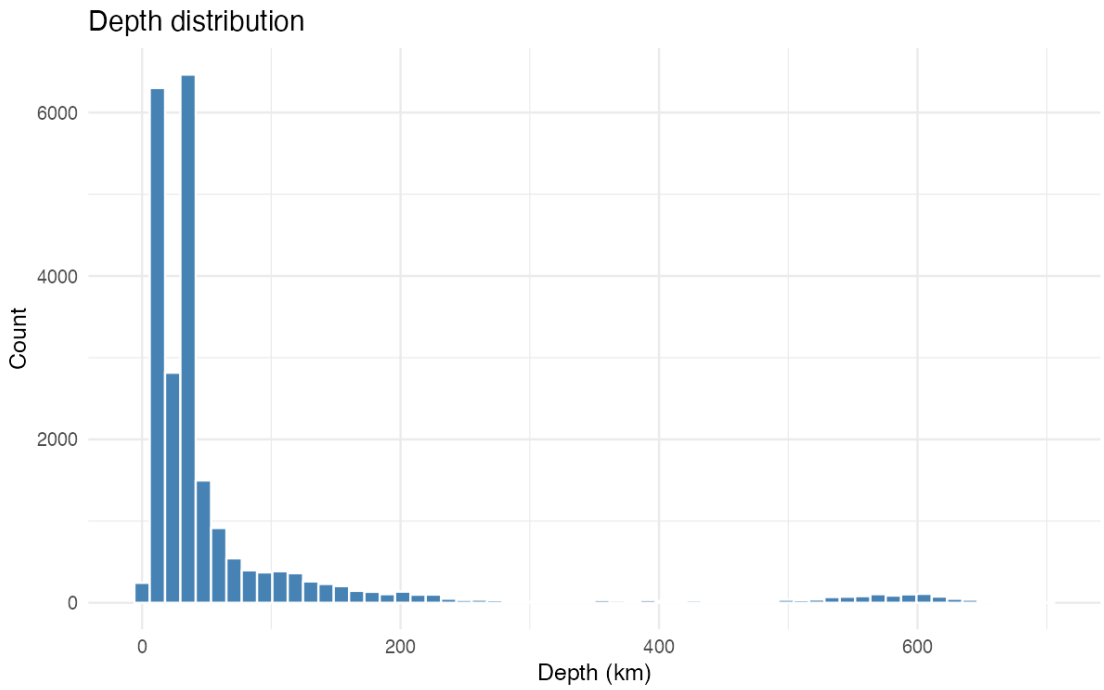


Figure 3: Distribution of earthquake depth (km).

Most earthquakes are shallow: 79.5% occur above 70 km and the median depth is 33 km. A smaller group lies in the intermediate range, and 1,326 events are deeper than 300 km (histogram above).

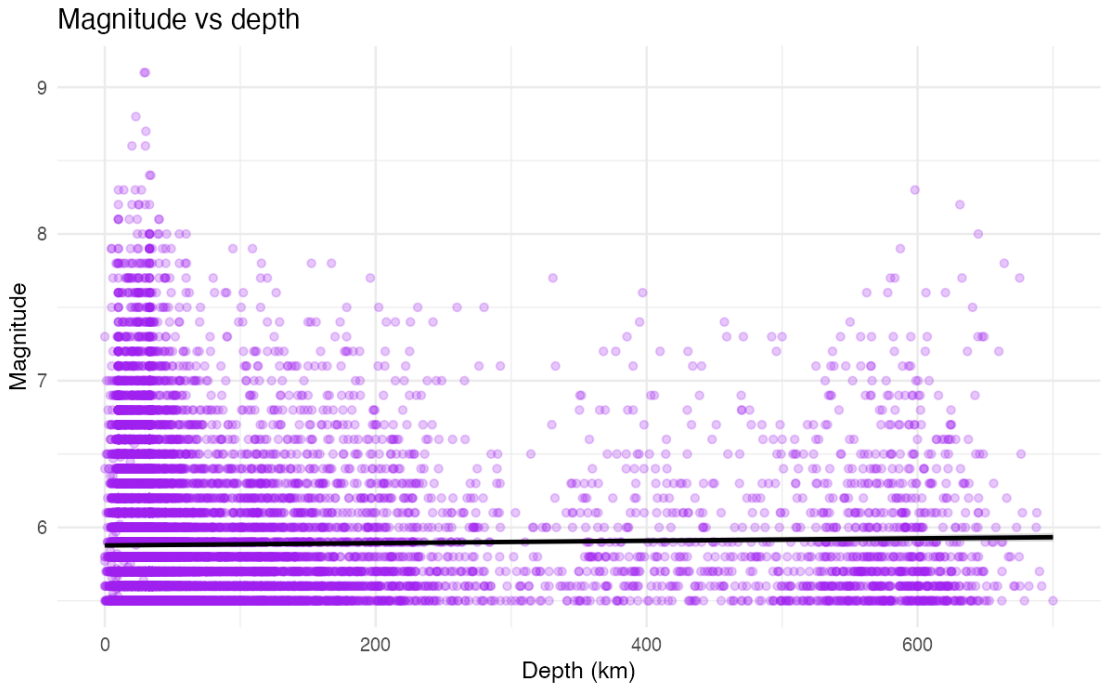


Figure 4: Magnitude against depth, with a linear trend line.

The scatter plot above shows almost no relationship between depth and magnitude. The Spearman correlation is -0.037 (p = 1.2e-08). With so many observations even a very small correlation

is statistically detectable, but it is practically negligible. A Kruskal–Wallis test finds a small but statistically significant difference in magnitude between depth categories (p = 2e-04).

5.2 Correlation

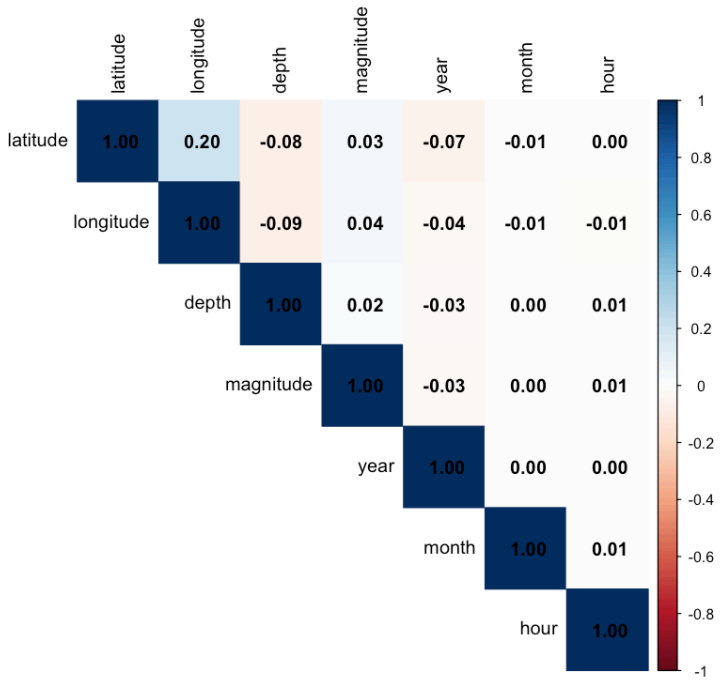


Figure 5: Correlation matrix of latitude, longitude, depth, magnitude, year, month and hour.

The correlations between the numeric variables are all weak (matrix above). This tells us that magnitude and depth carry largely independent information, which is why both are needed in the risk score.

6 Geographic and temporal analysis

6.1 Temporal patterns

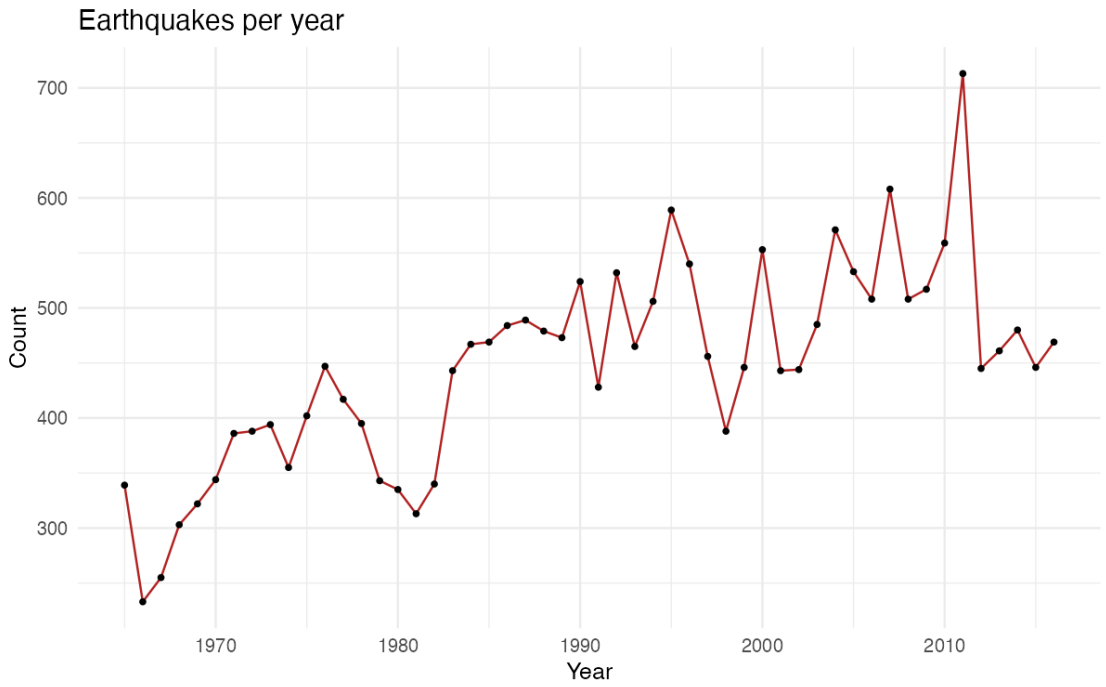


Figure 6: Number of earthquakes per year, 1965–2016.

The yearly count rises from the late 1960s to the 2000s (line chart above). A linear fit gives an increase of about 4.40 events per year (p = 1.3e-09). This should **not** be read as a real increase in seismic activity. Global catalogues become more complete and magnitudes are measured more consistently over time, so part of the rise is likely an artefact of detection and reporting. The busiest year is 2011 with 713 events. 253 of the 713 events in 2011 lie in a box around north-east Japan on or after 11 March 2011, which is consistent with the aftershock sequence of the magnitude 9.1 Tohoku earthquake.

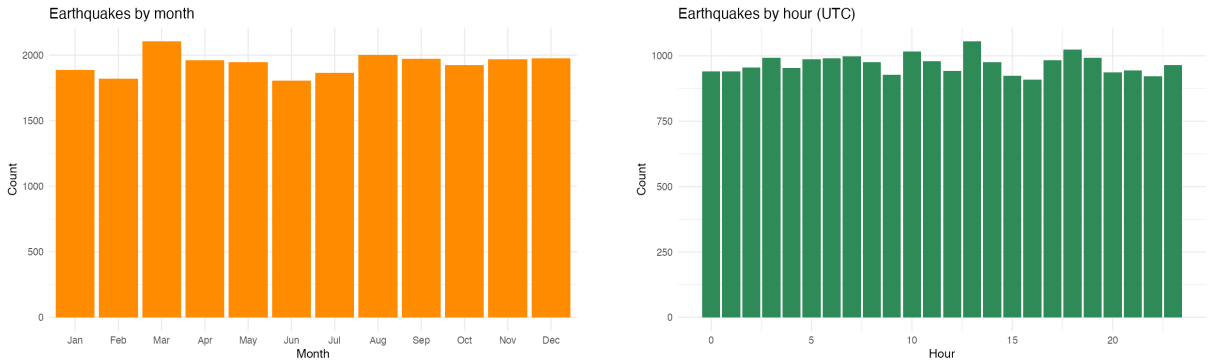


Figure 7: Earthquakes by calendar month (left) and by hour of day in UTC (right).

Counts by month and hour are fairly flat (bar charts above). A chi-square test rejects a perfectly uniform monthly distribution (p = 6.6e-05), but the test ignores month length and the fact that aftershocks are not independent events, so the small differences should not be interpreted as a

seasonal effect. There is no physical reason for a daily cycle, and none is visible.

6.2 Geographic patterns

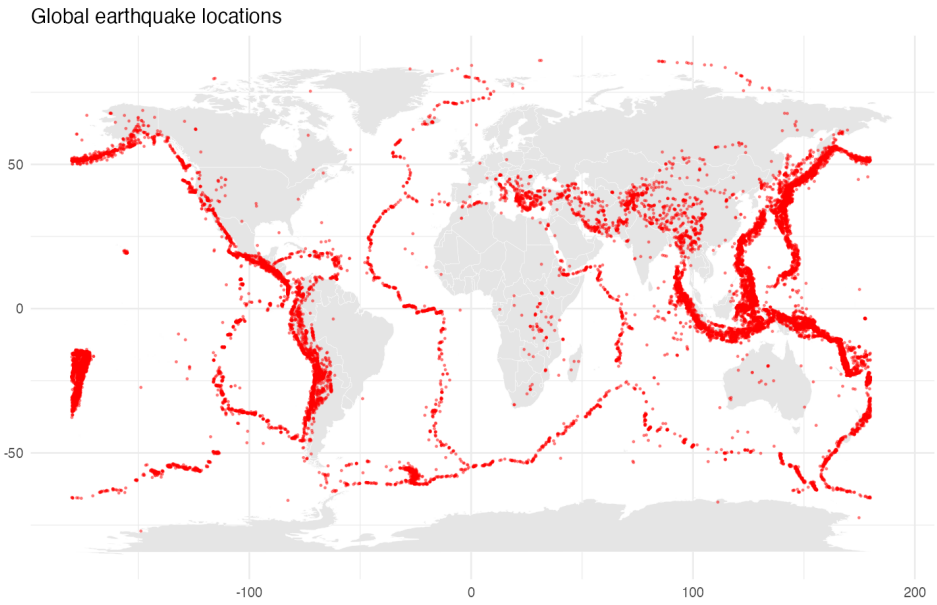


Figure 8: Epicentres of all earthquakes in the cleaned dataset.

The map above traces the plate boundaries: the Pacific "Ring of Fire", the Alpide belt from the Mediterranean to the Himalaya and Indonesia, and the mid-ocean ridges. Earthquakes are far from evenly spread over the globe.

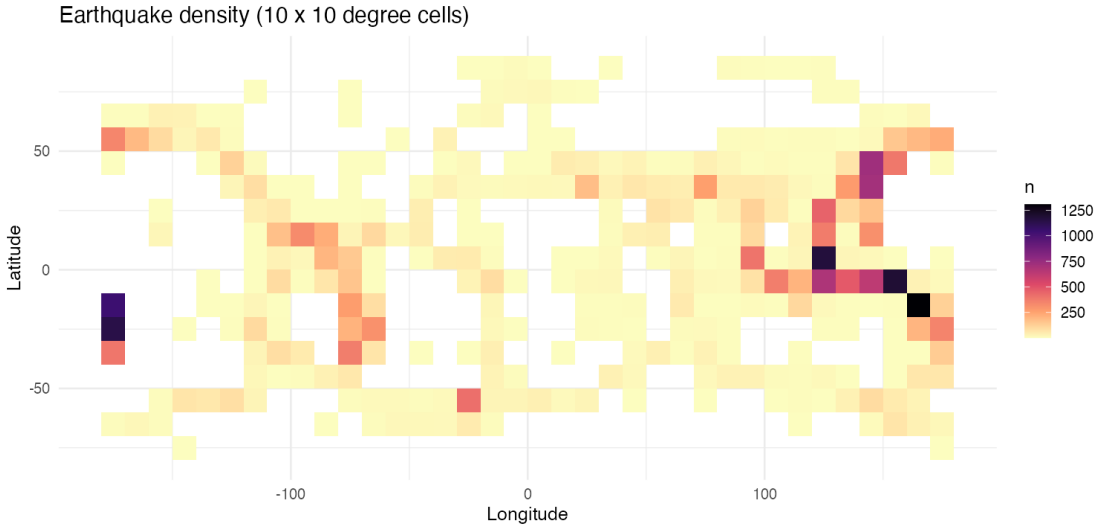


Figure 9: Earthquake counts in 10 x 10 degree grid cells (darker means more events).

Table 3: The eight most active 10 x 10 degree cells.

Latitude band	Longitude band	Earthquakes	Mean magnitude	Max magnitude
-20 to -10	160 to 170	1307	5.94	8.0
-10 to 0	150 to 160	1190	5.92	8.1

0 to 10	120 to 130	1171	5.87	8.0
-30 to -20	-180 to -170	1141	5.83	8.0
-20 to -10	-180 to -170	1043	5.84	8.1
40 to 50	140 to 150	727	5.92	8.3
30 to 40	140 to 150	716	5.89	9.1
-10 to 0	120 to 130	683	5.86	8.2

Table 3 confirms that the most active cells lie in the south-west and western Pacific, and that the busiest cell holds 1,307 events. Cell boundaries are arbitrary and not equal in area, so counts are only a rough guide to relative activity.

6.3 The strongest earthquakes

Table 4: The ten strongest earthquakes in the dataset.

Date (UTC)	Latitude	Longitude	Magnitude	Depth (km)	Risk level
2004-12-26	3.30	95.98	9.1	30.0	Severe
2011-03-11	38.30	142.37	9.1	29.0	Severe
2010-02-27	-36.12	-72.90	8.8	22.9	Severe
1965-02-04	51.25	178.72	8.7	30.3	Severe
2005-03-28	2.08	97.11	8.6	30.0	Severe
2012-04-11	2.33	93.06	8.6	20.0	Severe
2001-06-23	-16.26	-73.64	8.4	33.0	Severe
2007-09-12	-4.44	101.37	8.4	34.0	Severe
1994-10-04	43.77	147.32	8.3	14.0	Severe
2003-09-25	41.81	143.91	8.3	27.0	Severe

Table 4 lists the largest events. All of the ten strongest events are classed Severe or High, which follows directly from the way the score is built.

7 Risk score and risk categories

7.1 Definition

Each earthquake receives a score from 0 to 100:

$$\text{risk score} = 100 \times \left(0.6 \times \frac{M - 5.5}{M_{max} - 5.5} + 0.4 \times \left(1 - \frac{\min(D, 300)}{300} \right) \right)$$

where M is the magnitude, M_{max} the largest magnitude in the data and D the depth in km. Magnitude receives the larger weight (60%) because it measures the energy released. Shallowness receives 40% because shallow events usually shake the surface more strongly. Depth beyond 300 km is capped. The weights are a design choice made by the author for this study, not values derived from engineering data, and other reasonable weights would give different rankings.

Scores are grouped into four levels: **Low** (score up to 30), **Moderate** (above 30 to 42), **High** (above 42 to 55) and **Severe** (above 55).

Table 5: Earthquakes by risk level.

Risk level	Earthquakes	Share	Mean magnitude	Mean depth (km)
Low	3,375	14.5%	5.78	299.0
Moderate	11,877	51.1%	5.66	37.8
High	6,604	28.4%	6.11	25.0
Severe	1,376	5.9%	6.99	24.6

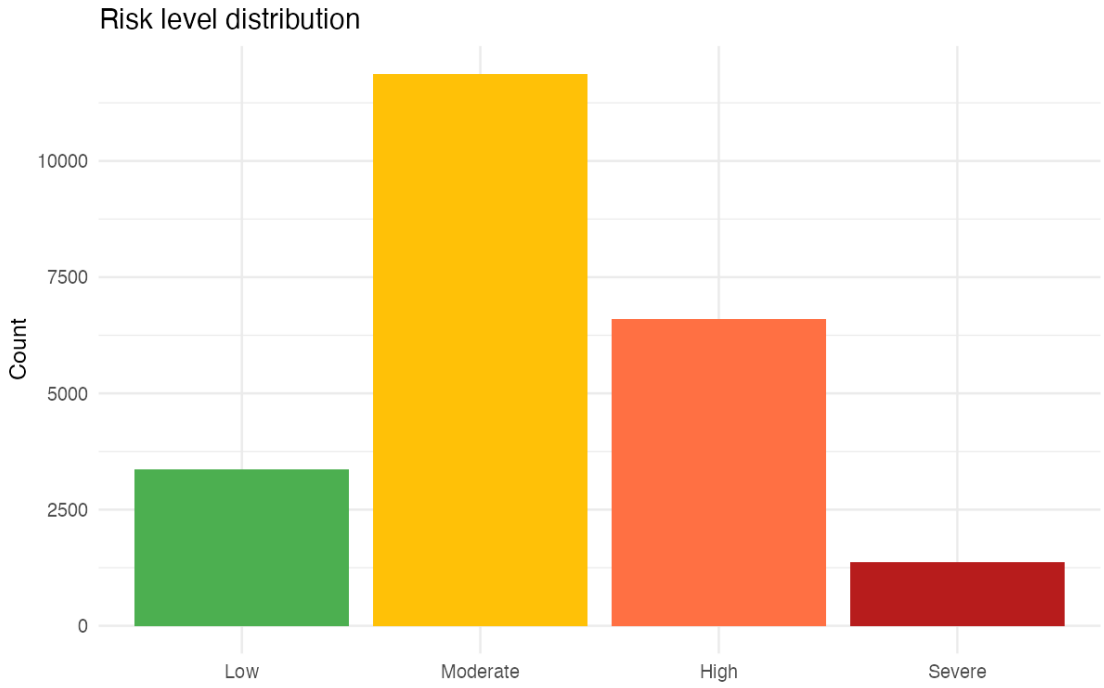


Figure 10: Number of earthquakes in each risk level.

Moderate is the largest level (51.1%), followed by High (28.4%). Severe events are rare (5.9%, 1,376 events). No Low event is shallow: of 3,375 Low events, 2,068 are at intermediate depth (70–300 km) and 1,307 are deeper than 300 km. A chi-square test confirms that risk level and depth category are strongly related ($p < 2.2e\text{-}16$), which is expected because depth is part of the score.

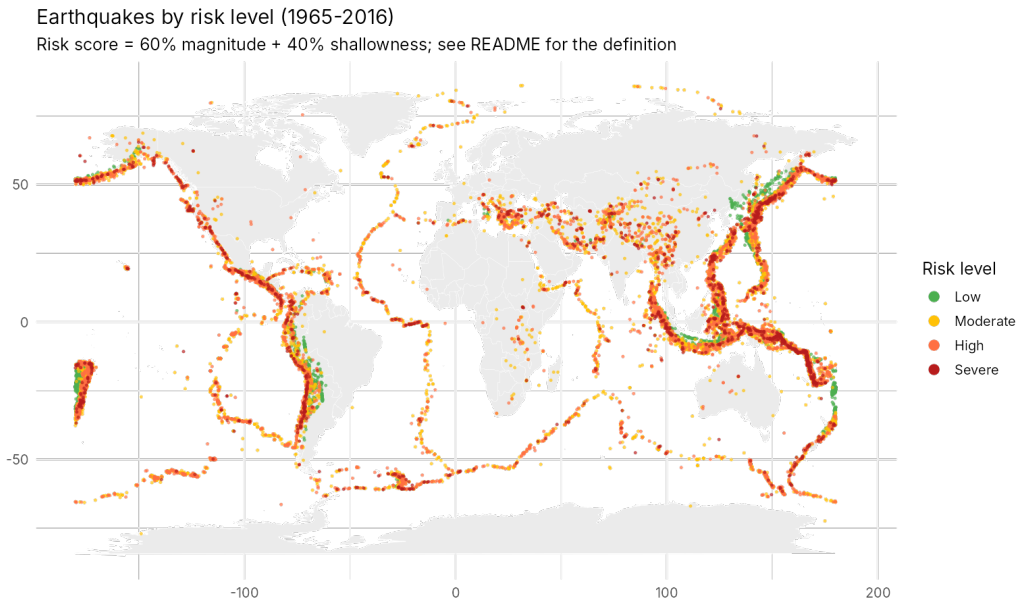


Figure 11: Earthquakes coloured by risk level; Severe events are drawn on top.

Severe and High events concentrate along subduction zones of the western and south-western Pacific and the South American coast (map above).

7.2 Risk by region and over time

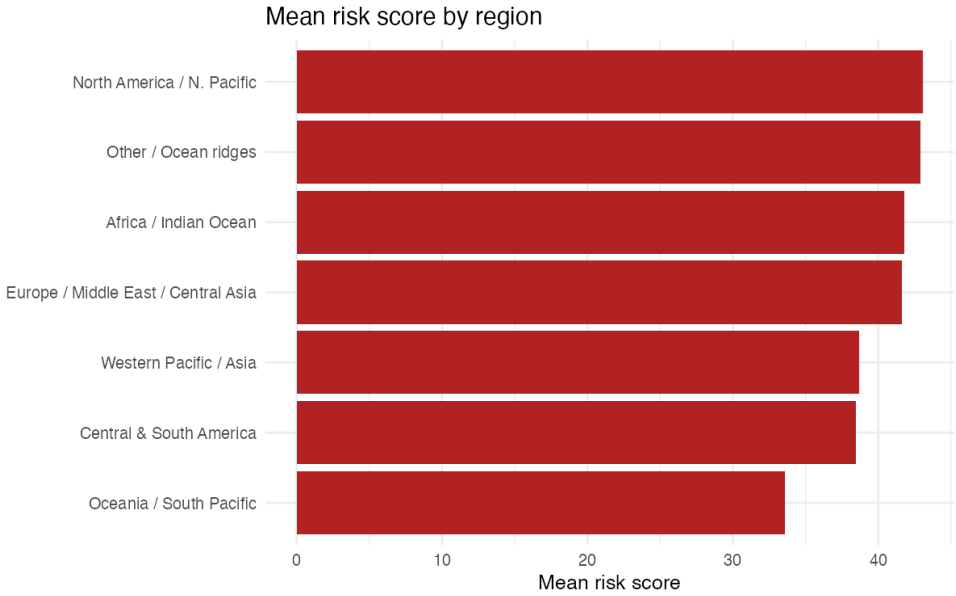


Figure 12: Mean risk score by region.

Table 6: Risk summary by region. Regions are rough latitude–longitude boxes.

Region	Earthquakes	Mean risk	% High/Severe
North America / N. Pacific	1,340	43.1	47.0
Other / Ocean ridges	1,456	42.9	43.5
Africa / Indian Ocean	1,229	41.8	37.7
Europe / Middle East / Central Asia	1,458	41.6	40.1

Western Pacific / Asia	10,994	38.7	32.9
Central & South America	2,619	38.5	32.8
Oceania / South Pacific	4,136	33.6	28.7

Table 6 shows only modest differences in mean score between regions (from 33.57 to 43.05). The regions are crude bounding boxes defined in the script and the "Other / Ocean ridges" group is a catch-all, so this table is a first look rather than a tectonic classification.

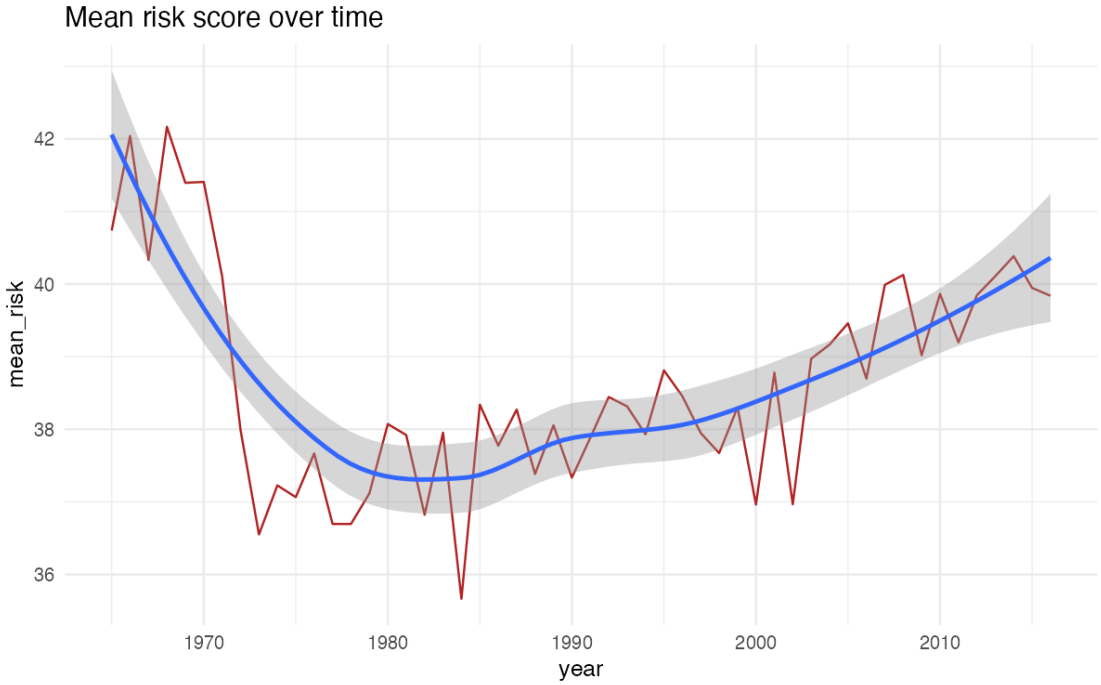


Figure 13: Mean risk score per year with a smoothed trend.

8 Clustering analysis

K-means was applied to five variables: latitude and longitude converted to three coordinates on a unit sphere (so that longitude wraps correctly at $\pm 180^{\circ}$), the standardised log depth and the standardised magnitude. The elbow and silhouette methods were run on a random sample of 4,000 earthquakes (plots below).

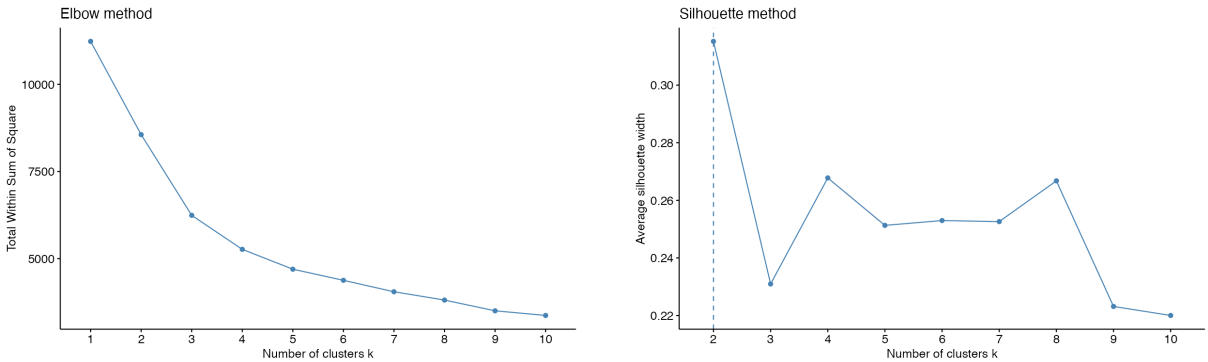


Figure 14: Elbow method (left) and average silhouette width (right) for k = 1 to 10.

The elbow curve decreases smoothly and has no sharp bend. The average silhouette width is highest for k = 2 (about 0.32); for k = 6 it is about 0.25. The analysis uses **k = 6** as a pragmatic choice that yields more detailed groups, but the diagnostics do not single out six clusters, so the clusters should be seen as a descriptive grouping rather than as natural, well separated classes.

Table 7: Cluster profile (K-means, k = 6).

Cluster	Earthquakes	Mean magnitude	Mean depth (km)	Mean risk score
1	8,186	5.67	41.2	37.3
2	2,368	5.75	64.5	35.6
3	3,796	6.28	25.4	49.6
4	1,308	7.10	86.4	57.3
5	2,795	5.83	324.6	15.5
6	4,779	5.70	10.4	41.9

Table 7 shows that the clusters differ mainly by depth and magnitude. Cluster 5 collects the deep events (mean depth 325 km, lowest mean risk 15.5), cluster 4 the large events (mean magnitude 7.10, highest mean risk 57.3), and cluster 6 the shallowest events (mean depth 10.4 km). The mean latitude and longitude of a cluster are not meaningful locations because several clusters are spread across the whole globe.

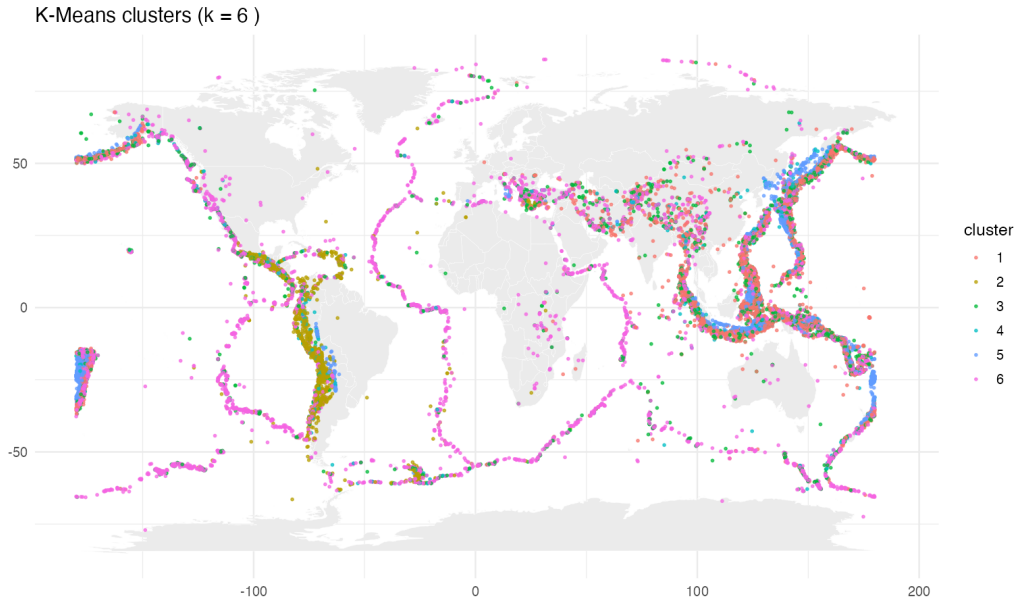


Figure 15: K-means clusters (k = 6) on a world map.

In the map above the clusters overlap strongly in space along the plate boundaries because depth and magnitude, not only position, drive the grouping. The one visible regional pattern is that events along the western coast of the Americas and the Caribbean are largely assigned to a single cluster (cluster 2).

9 Machine-learning models and evaluation

9.1 Setup

The goal is a binary classification: is an earthquake **High/Severe** (the positive class, 34.3% of the data) or **Low/Moderate**? Magnitude and depth were deliberately *excluded* from the inputs because they define the risk score, and including them would make the task trivial (data leakage). The models therefore use latitude, longitude, year, month, hour, review status and magnitude type. The data were split 80/20 with stratification (about 4,650 test earthquakes). Three models were trained: logistic regression, a decision tree (`rpart`, cp = 0.005) and a random forest (300 trees). Because the classes are imbalanced, accuracy alone is misleading: a model that always predicts Low/Moderate would reach an accuracy of 0.66.

9.2 Results

Table 8: Test-set performance. The positive class is High/Severe.

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.664	0.561	0.106	0.178	0.647
Decision Tree	0.660	0.526	0.112	0.185	0.616
Random Forest	0.692	0.592	0.336	0.428	0.722



Figure 16: Model comparison on the test set.

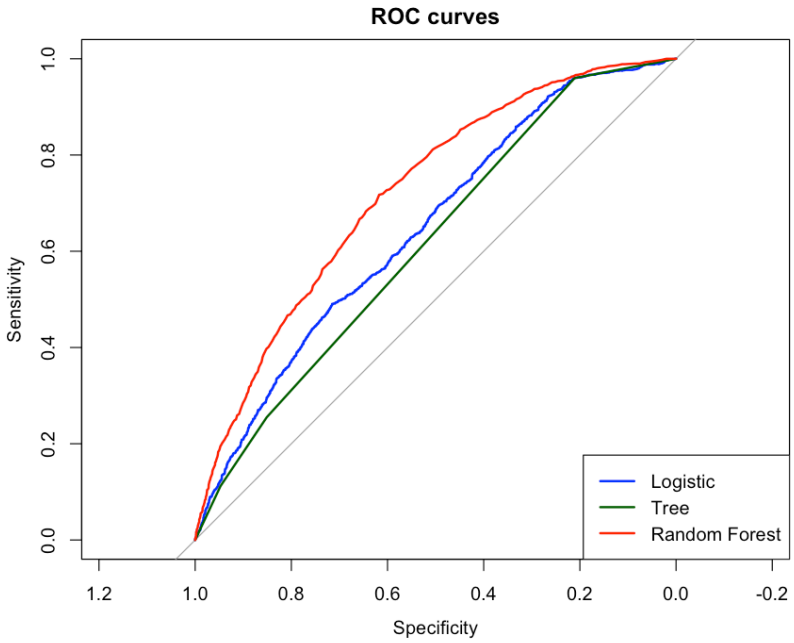


Figure 17: ROC curves of the three models on the test set.

The random forest is the best model on every metric (Table 8 and the two figures above): accuracy 0.692, precision 0.592, recall 0.336, F1 0.428 and AUC 0.722. Logistic regression and the decision tree have an accuracy (0.664 and 0.660) that is barely above the majority-class baseline of 0.657, and they detect only about 11% of High/Severe events (recall 0.11 and 0.11). The random forest finds about 34% of them. In the ROC plot the random forest curve lies above the other two, with an AUC of 0.72, which indicates moderate ability to rank High/Severe events above others. Overall, location, time and catalogue information carry some, but limited, information about whether an event is High/Severe.

9.3 Decision tree and feature importance

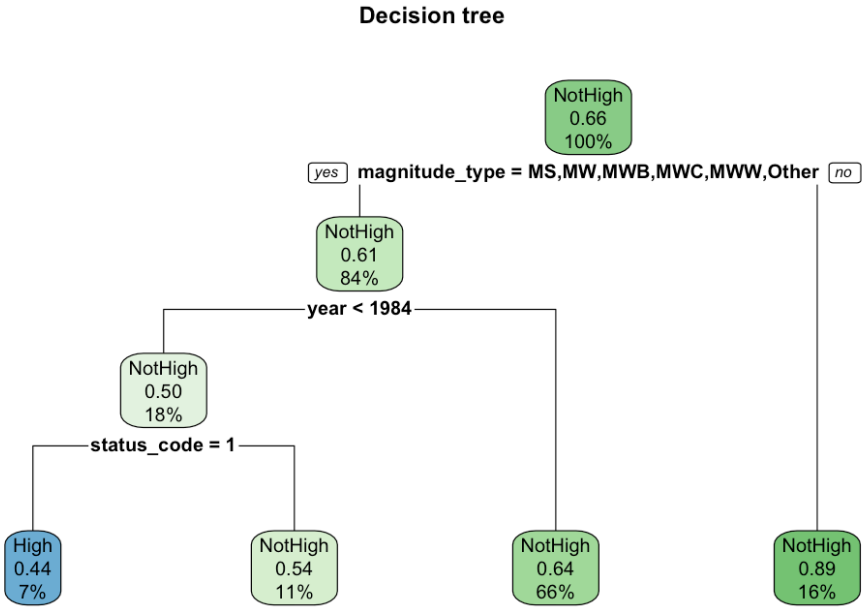


Figure 18: Decision tree (cp = 0.005).

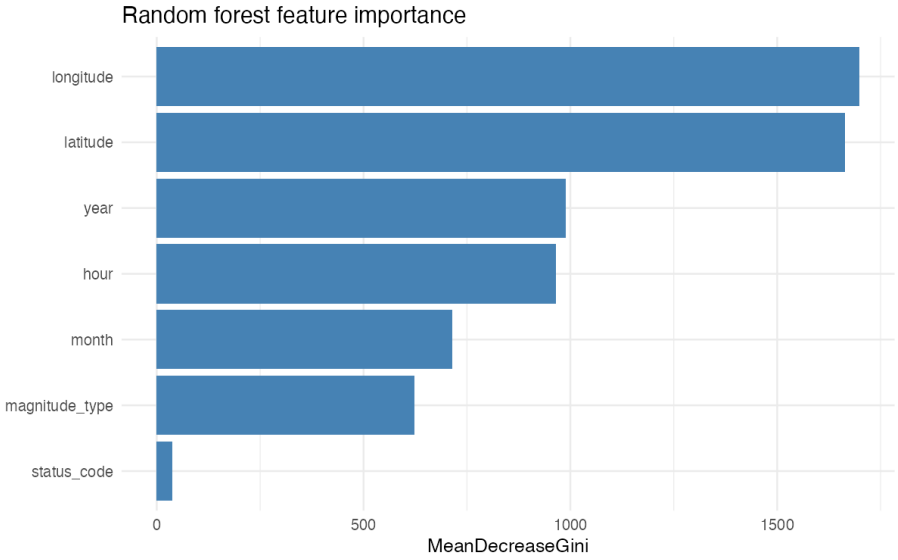


Figure 19: Random forest feature importance (mean decrease in Gini).

Table 9: Random forest variable importance.

Feature	Mean decrease accuracy	Mean decrease Gini
longitude	47.9	1698.9
latitude	48.1	1664.7
year	51.7	988.8
hour	9.1	965.9
month	6.7	714.2
magnitude_type	83.2	622.4

status_code	21.2	38.0

Longitude and latitude have the largest Gini importance (figure above and Table 9), reflecting that High/Severe events concentrate along particular plate boundaries. `year` and `magnitude_type` are also important. These two describe how an event was catalogued rather than a physical cause: the magnitude scale used is related to the size of an event, and the catalogue changes over the years. They therefore act partly as indirect proxies for magnitude, and the model should not be read as discovering physical drivers of risk.

10 Dashboard

The interactive Shiny dashboard (`app.R`) lets a user explore the same cleaned dataset used in this report. It offers:

- **Filters** (sidebar): magnitude, depth, year range, risk level, region and K-means cluster, with a reset button.
- **Overview:** key figures (count, mean and maximum magnitude, median depth, share of High/Severe events) and interactive magnitude, yearly, depth and magnitude-versus-depth charts.
- **Risk analysis:** risk-level distribution, mean risk per year and a regional summary table.
- **Map:** an interactive map coloured by magnitude or by risk level (a random sample of up to 5,000 events is drawn for speed).
- **Clusters:** a cluster profile table that updates with the filters.
- **Models:** the saved model-comparison table, the best model and the random forest feature importance (these are fixed results, independent of the filters).
- **Data:** the filtered records in a searchable table with CSV download.
- **About:** the risk-score definition and data notes.

When a filter combination matches no earthquakes the app shows a short message instead of an error, and the Models tab shows a notice if the saved result files are missing. The dashboard runs locally with `shiny::runApp()`; the README gives the exact steps and the instructions for publishing it on shinyapps.io. At the time of writing, the public deployment link is listed in the README only once the app has been published and checked.

11 Limitations and future work

Limitations

- *Descriptive score.* The risk score is a simple formula with weights chosen by the author. It ignores population density, building quality, ground conditions and tsunami potential.
- *Catalogue effects.* The dataset only contains events of magnitude 5.5 and above, and the rise in yearly counts is probably influenced by improving detection and reporting.
- *Clusters.* The number of clusters (k = 6) is not strongly supported by the elbow and silhouette diagnostics (best silhouette at k = 2).
- *Model skill and indirect leakage.* The classifiers are only moderately better than the baseline, and inputs such as `year` and `magnitude_type` act partly as proxies for magnitude. The models classify recorded events; they do not predict future earthquakes.

- *Sensitivity of results.* A repeat run of the pipeline in a different R environment (R 4.3.3) reproduced the data cleaning, the risk-level counts and the Spearman correlation exactly, and gave the same ranking of models (random forest best, F1 about 0.43, AUC about 0.72). However, random steps (data split, clustering) gave different values, and in that run the decision tree collapsed to a single node. The decision-tree results in this report should therefore be treated as sensitive to the random split, and a repeated or cross-validated evaluation would be more reliable.
- *Missing outputs.* The original run did not save confusion matrices. The updated analysis script now writes them to `outputs/tables/confusion_matrices.csv` and `outputs/plots/63_confusion_matrices.png`; they are not shown here because they were not part of the run that produced the reported metrics.
- *Regions.* The region labels are coarse latitude–longitude boxes.

Future work

- Use cross-validation and a fixed training/test partition saved to disk so that all metrics are reproducible exactly.
- Tune the models and compare against gradient boosting; handle class imbalance (for example class weights or threshold tuning).
- Choose the number of clusters with more care, and test density-based clustering such as DBSCAN on the sphere.
- Add plate-boundary and population data to build a more realistic exposure-based risk measure.
- Publish the dashboard and add screenshots to the repository.

12 Conclusion

Starting from a public catalogue of significant earthquakes, this study cleaned 23,412 raw records to 23,232 natural earthquakes, described their magnitude, depth, temporal and geographic patterns, and built a transparent risk score that places about 34.3% of events in the High or Severe levels. Most events are shallow (79.5% above 70 km) and cluster along plate boundaries, especially in the western and south-western Pacific. K-means grouped events mainly by depth and magnitude, with the caveat that the choice of six clusters is not strongly supported. A random forest recognised High/Severe events best (F1 0.43, AUC 0.72) but only moderately better than a trivial baseline, which shows how much of the risk depends on magnitude and depth themselves. The complete workflow is reproducible from the public repository, and the Shiny dashboard makes the results easy to explore.

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